

TẦNG 1

"Presentation Layer" {

```
class MainMenu <<boundary>> {  
    + submit_login_form(): void  
    + submit_register_form(): void  
}
```

```
class AdminUI <<page>> {  
    + view_dashboard_metrics(): void  
    + manage_movie(): void  
    + manage_showtime(): void  
    + manage_room(): void  
    + config_display(): void  
    + sell_ticket_offline(): void  
    + view_statistics(): void  
    + cancel_ticket(): void  
}
```

```
class CustomerUI <<page>> {  
    + user_obj: UserData  
    + selected_movie: String  
    + selected_seats: List<String>  
    + payment_step: bool  
    + ad_closed: bool  
    + current_page: String  
    + target_date: String  
    + target_time: String  
    + generated_ticket_ids: List<String>  
    + payment_start_time: double  
  
    + view_home(): void  
    + quick_booking(): void  
    + select_seat(): void  
    + process_payment(): void  
    + book_ticket(): void  
    + view_history(): void  
}
```

// Mối quan hệ phụ thuộc (Dependency) gọi xuống Tầng 2

MainMenu ..> AuthController : <<use>>

AdminUI ..> AdminController : <<use>>

AdminUI ..> MovieController : <<use>>

```

AdminUI ..> ShowtimeController : <<use>>
AdminUI ..> RoomController : <<use>>
CustomerUI ..> BookingController : <<use>>
CustomerUI ..> MovieController : <<use>>
CustomerUI ..> ShowtimeController : <<use>>

```

```

// Các hàm giao diện tự do phụ trợ (từ ui_components.py)
+ navigate_to(page: String, movie: String = ""): void
+ show_advertisement(): void
+ show_popcorn_effect(): void
+ create_premium_movie_card(col: Any, title: String, genre: String, duration: int,
base_price: double, poster_path: String): void
}

```

TẦNG 2

```

"Business Logic Layer" {

```

```

class AuthController {
- _io_handler: FileIOHandler
- _user_table: UserHashTable
- _current_user: UserData
- _auth_lock: Lock

- _generate_user_id(): String
+ login(username: String, pass: String): String
+ register(username: String, pass: String, confirm_password: String): bool
+ logout(): void
+ get_current_user(): UserData
+ is_logged_in(): bool
+ is_admin(): bool
+ get_user_table(): UserHashTable
+ get_all_users(): List<UserData>
}

```

```

class AdminController {
- _movie_controller: MovieController
- _booking_controller: BookingController

+ calculate_revenue(): double
+ count_movies(): int
+ sell_ticket_at_counter(movie: MovieData, showtime: Showtime, row: int, col: int):
bool/String
+ generate_report(): Map<String, Any>
}

```

```
+ get_active_tickets(): List<TicketData>
}
```

```
class MovieController {
- _io_handler: FileIOHandler
- _movie_list: MovieLinkedList

+ generate_movie_id(): String
+ add_movie(movie: MovieData): bool
+ search_by_id(movie_id: String): MovieNode
+ search_by_title(title: String): MovieNode
+ delete_movie(movie_id: String): bool
+ get_movie_list(): MovieLinkedList
+ get_movie_data(): List<MovieData>
+ movie_exists(movie_id: String): bool
+ save_data(): void
+ load_ui_config(): List<List<String>>
+ save_ui_config(slider_titles: List<String>, list_titles: List<String>): void
}
```

```
class BookingController {
- _io_handler: FileIOHandler
- _showtime_controller: ShowtimeController
- _movie_controller: MovieController
- _room_controller: RoomController
- _ticket_list: TicketLinkedList
- _booking_lock: Lock

- _parse_seat_id(seat_id: String): Pair<int, int>
- _sync_matrix_from_tickets(): void
+ get_booking_history(user_id: String): List<TicketData>
+ get_ticket_data(): List<TicketData>
+ get_ticket_list(): TicketLinkedList
+ refresh_booking_data(): void
}
```

```
class ShowtimeController {
- _io_handler: FileIOHandler
- _showtime_list: ShowtimeLinkedList

+ generate_showtime_id(): String
+ get_showtime_list(): ShowtimeLinkedList
+ get_showtime_data(): List<Showtime>
+ add_showtime(showtime: Showtime): bool
}
```

```

+ get_unique_dates(movie_id: String): List<String>
+ get_unique_times(movie_id: String, target_date: String): List<String>
+ get_showtimes_by_movie(movie_id: String): List<Showtime>
+ find_exact_showtime(movie_id: String, target_date: String, target_time: String):
Showtime
+ extract_date(st: Showtime): String
+ extract_time(st: Showtime): String
}

```

```

class RoomController {
- _io_handler: FileIOHandler
- _room_list: RoomLinkedList

+ add_room(room: Room): bool
+ find_room(room_id: String): RoomNode
+ find_room_by_name(room_name: String): Room
+ get_room_list(): RoomLinkedList
+ get_room_data(): List<Room>
+ count_rooms(): int
}

```

// Mối quan hệ tương tác nội bộ giữa các bộ điều khiển (Internal Layer Relationships)

AdminController --> MovieController : <<call>>

AdminController --> BookingController : <<call>>

BookingController --> ShowtimeController : <<call>>

BookingController --> MovieController : <<call>>

BookingController --> RoomController : <<call>>

// Mối quan hệ phụ thuộc gọi xuống Tầng 3 (Data Structures) và Tầng 4 (Data Models)

AdminController ..> UserHashTable : <<manage>>

MovieController ..> MovieLinkedList : <<manage>>

BookingController ..> TicketLinkedList : <<manage>>

ShowtimeController ..> ShowtimeLinkedList : <<manage>>

RoomController ..> RoomLinkedList : <<manage>>

AdminController ..> FileIOHandler : <<use>>

MovieController ..> FileIOHandler : <<use>>

BookingController ..> FileIOHandler : <<use>>

ShowtimeController ..> FileIOHandler : <<use>>

RoomController ..> FileIOHandler : <<use>>

```

}

```

TẦNG 3

TẦNG 3

"Data Structures Layer" {

```
class Node {  
    + data: Any  
    + next: Node  
  
    + get_data(): Any  
    + set_data(data: Any): void  
    + get_next(): Node  
    + set_next(next_node: Node): void  
}
```

```
class UserNode <<node>> {}  
class TicketNode <<node>> {}  
class MovieNode <<node>> {}  
class ShowtimeNode <<node>> {}  
class RoomNode <<node>> {}
```

```
class UserHashTable {  
    - _SIZE: int  
    - _table: List<UserNode>  
  
    + hash_function(key: String): int  
    + insert(username: String, value: UserData): void  
    + get(username: String): UserData  
    + delete(username: String): bool  
    + display(): void  
    + get_table(): List<UserNode>  
    + get_all(): List<UserData>  
}
```

```
class Array2D {  
    + rows: int  
    + cols: int  
    + data: List<Any>  
  
    + get_val(r: int, c: int): Any  
    + set_val(r: int, c: int, value: Any): void  
}
```

```
class TicketLinkedList {  
    - _head: TicketNode  
    - _tail: TicketNode
```

```

+ add_ticket(ticket: TicketData): void
+ find_ticket(ticket_id: String): TicketNode
+ remove_ticket(ticket_id: String): bool
+ get_head(): TicketNode
+ get_tail(): TicketNode
}

```

```

class MovieLinkedList {
- _head: MovieNode

+ add_movie(movie: MovieData): void
+ search_id(movie_id: String): MovieNode
+ search_title(title: String): MovieNode
+ remove_movie(movie_id: String): bool
+ get_head(): MovieNode
}

```

```

class ShowtimeLinkedList {
- _head: ShowtimeNode

+ add_showtime(showtime: Showtime): void
+ find_by_movie(movie_id: String): List<Showtime>
+ get_head(): ShowtimeNode
}

```

```

class RoomLinkedList {
- _head: RoomNode

+ add_room(room: Room): void
+ find_room(room_id: String): RoomNode
+ remove_room(room_id: String): bool
+ get_head(): RoomNode
+ traverse(): void
}

```

// Quan hệ kế thừa (Inheritance) từ lớp cơ sở Node

```

UserNode --|> Node
TicketNode --|> Node
MovieNode --|> Node
ShowtimeNode --|> Node
RoomNode --|> Node

```

// Quan hệ chứa đựng và quản lý các Node nội bộ (Composition / Aggregation)

```

UserHashTable "1" *--> "0..*" UserNode : <<manage>>
TicketLinkedList "1" *--> "0..*" TicketNode : <<manage>>
MovieLinkedList "1" *--> "0..*" MovieNode : <<manage>>
ShowtimeLinkedList "1" *--> "0..*" ShowtimeNode : <<manage>>
RoomLinkedList "1" *--> "0..*" RoomNode : <<manage>>
}

```

TÀNG 4

```
"Data Models Layer" {
```

```

class FileIOHandler {
    + base_path: String
    + users_file: String
    + movies_file: String
    + rooms_file: String
    + showtimes_file: String
    + tickets_file: String

    + load_users(table: UserHashTable): void
    + save_users(table: UserHashTable): void
    + load_movies(movie_list: MovieLinkedList): void
    + save_movies(movie_list: MovieLinkedList): void
    + load_rooms(room_list: RoomLinkedList): void
    + save_rooms(room_list: RoomLinkedList): void
    + load_showtimes(showtime_list: ShowtimeLinkedList): void
    + save_showtimes(showtime_list: ShowtimeLinkedList): void
    + load_tickets(ticket_list: TicketLinkedList): void
    + save_tickets(ticket_list: TicketLinkedList): void
    + load_ui_config(): List<List<String>>
    + save_ui_config(slider_titles: List<String>, list_titles: List<String>): void
    + loadData(users: UserHashTable, movies: MovieLinkedList, rooms: RoomLinkedList,
showtimes: ShowtimeLinkedList, tickets: TicketLinkedList): void
    + saveData(users: UserHashTable, movies: MovieLinkedList, rooms: RoomLinkedList,
showtimes: ShowtimeLinkedList, tickets: TicketLinkedList): void
}

```

```

class UserData <<entity>> {
    - _username: String
    - _password: String
    - _role: String
    - _user_id: String

    + get_username(): String

```

```
+ get_password(): String
+ get_role(): String
+ get_user_id(): String
+ check_password(password: String): bool
}
```

```
class MovieData <<entity>> {
  - _movie_id: String
  - _title: String
  - _genre: String
  - _duration: int
  - _description: String
  - _base_price: double
  - _poster_path: String
  - _revenue: double

  + get_movie_id(): String
  + get_title(): String
  + get_genre(): String
  + get_duration(): int
  + get_description(): String
  + get_base_price(): double
  + get_poster_path(): String
  + get_revenue(): double
  + set_title(title: String): void
  + set_genre(genre: String): void
  + set_duration(duration: int): void
  + set_description(description: String): void
  + set_base_price(base_price: double): void
  + set_poster_path(poster_path: String): void
  + add_revenue(amount: double): void
}
```

```
class Room <<entity>> {
  - _room_id: String
  + room_name: String
  + rows: int
  + cols: int
  - _capacity: int

  + get_room_id(): String
  + get_room_name(): String
  + get_rows(): int
  + get_cols(): int
}
```



```
+ get_capacity(): int  
}
```

```
class SeatMatrix {  
  - _rows: int  
  - _cols: int  
  - _seats: Array2D  
  
  + generate_seat_id(r: int, c: int): String  
  + check_status(r: int, c: int): int  
  + reserve_seat(r: int, c: int): bool  
  + book_seat(r: int, c: int): bool  
  + release_seat(r: int, c: int): void  
  + get_seats(): Array2D  
  + load_matrix(seats: List<Any>): void  
  + set_seat_status(r: int, c: int, status: int): bool  
  + get_rows(): int  
  + get_cols(): int  
}
```

```
class Showtime <<entity>> {  
  - _showtime_id: String  
  - _movie_id: String  
  - _start_time: String  
  - _room_id: String  
  - _seat_matrix: SeatMatrix  
  
  + get_showtime_id(): String  
  + get_movie_id(): String  
  + get_room_id(): String  
  + get_start_time(): String  
  + get_seat_matrix(): SeatMatrix  
  + get_available_seats(): int  
}
```

```
class TicketData <<entity>> {  
  - _ticket_id: String  
  + _user_id: String  
  + _movie_id: String  
  + _seat_id: String  
  + _status: String  
  + _showtime_id: String  
  + _room_id: String  
  + _price: double
```

```

+ _booking_time: String

+ get_ticket_id(): String
+ get_user_id(): String
+ get_movie_id(): String
+ get_seat_id(): String
+ get_status(): String
+ get_showtime_id(): String
+ get_room_id(): String
+ get_price(): double
+ get_booking_time(): String
}

// Quan hệ cấu thành (Composition) trong tầng thực thể
Showtime "1" *--> "1" SeatMatrix : <<contains>>

// Quan hệ phụ thuộc từ FileIO xử lý các thực thể
FileIOHandler ..> UserData : <<serialize>>
FileIOHandler ..> MovieData : <<serialize>>
FileIOHandler ..> Room : <<serialize>>
FileIOHandler ..> Showtime : <<serialize>>
FileIOHandler ..> TicketData : <<serialize>>
}

```